



Review

The ‘training load’ construct: Why it is appropriate and scientific

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ABSTRACT

A recent paper called for the abandonment of the term load (and training load) when used outside its mechanical meaning, claiming it is “unscientific” and “breaches scientific principles.” In this article, we explain why its use does not breach any scientific principles and we clarify the process of labelling, conceptualising and operationalising a construct. Training load is simply a label attributed to a higher-order construct overarching other interrelated sub-dimensions. This multi-level structure provides a framework (nomological network) to support the research process and also practical applications. Load is a word, and therefore cannot be “unscientific”. The “use” or “misuse” of words and terms entirely depends upon definitions that should be based on current understanding. Misuse occurs when a term is decontextualised or interpreted according to a unilateral perspective. The field of mechanics does not have a monopoly on the term load (or other common terms such as work, stress and fatigue), which are legitimately used in many scientific areas and with various meanings. The ‘obligation’ to rely on terms abiding by the *Système International d’Unités* (SI) when describing a construct is inappropriate. The SI relates to how we can measure, not describe training load; i.e., SI is relevant to its operational and not its constitutive (descriptive) definition. Discussions regarding shared and standardised descriptions and definitions are more relevant than discussions about discarding terms in sport and exercise science. Researchers (and practitioners) can continue to use the term training load as it does not breach any scientific principles.

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1. Introduction

Recently, an opinion piece was published in this journal that suggested dismissing the use of the term *load* (and hence *training load*) when it is used outside of a mechanical context, claiming it is “unscientific” and “breaches principles of science”.¹ Whilst the call for the better use of definitions is welcome, within this article we explain why the use of the term *load* in *training load* does not breach any scientific principles, and why it can be used productively in both practice and research. To aid our explanation, we will preface some arguments by referring to certain misconceptions presented within the article by Staunton and colleagues¹ that are used as foundations for constructive discussion and education. Additionally, to assist with future clarity and understanding on this topic, we provide a detailed explanation of constructs (such as *training load*), laying out what they are and how to

appropriately label, conceptualise and operationalise them so that they can be investigated using the scientific method. A more detailed rebuttal to the contents of the article by Staunton and colleagues,¹ on which this article is based, and that contains the complete reference list, has been published as a preprint (<https://sportrxiv.org/index.php/server/preprint/view/16>).

2. Is the use of *load* in *training load* “unscientific”?

While it has been proposed that “the term ‘training load’ is unscientific and must therefore be abandoned”,¹ it should be noted that words or terms by themselves cannot be considered unscientific. For example, the word (and construct) *spiritualism* may, at first glance, appear to be “unscientific.” However, this word can be defined and conceptualised in a way that it meets the requirements to be considered a scientific construct and investigated using the scientific method.² Something can be considered unscientific when it cannot be investigated following the scientific method. Indeed, *scientific* relates to the process of acquiring

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knowledge and is not simply a semantic property of a term. This is coherent with the definition of science provided by the Science Council (<https://sciencecouncil.org/>):

“Science is the pursuit and application of knowledge and understanding of the natural and social world following a systematic methodology based on evidence.”

Whilst words or terms by themselves cannot be deemed as ‘unscientific’, the misuse of words and terms can be classified as poor practice because it confounds mutual understanding, thus impairing scientific progress. Similarly, it is poor practice to extrapolate, decontextualise or reinterpret arguments and definitions according to unilateral and subjective perspectives.

The field of mechanics does not have a monopoly on the term “load” in science

While the term *load* is utilised in the field of mechanics, its use outside of this context cannot be considered ‘improper’¹ and this argument fails to consider the current understanding and use of this term in other valid settings. Indeed, many other constructs commonly use *load* outside of its mechanical meaning, such as *allostatic load*, *cognitive load*, *mental load*, *glycemic load*, *hemodynamic load*, or the now frequently used *viral load*. These are just some examples where *load* has other, legitimate, contextual connotations, the majority of which are relevant to exercise science. Exercise science is an interdisciplinary field that includes several sub-disciplines, of which biomechanics is just one. The idea that one specific definition or context should prevail over all others is unjustified, and there is no scientific principle stating that the field of mechanics should take precedence over all other scientific areas and contexts. The use of the term *load* has not limited the investigation of these constructs from a scientific point of view. Indeed, the aforementioned constructs utilising *load*, along with *training load*, are used in science, are defined, describe specific constructs and concepts, are measurable, and have been investigated. The field of mechanics does not have a monopoly on the term *load* (or other commonly used terms such as *work*, *stress*, and *fatigue*), which are legitimately used in many scientific areas with various meanings.

Terms and nomenclature used to describe exercise should abide by the Système International d’Unités: a misconception

A primary reasoning put forth to justify why the use of the term *load* outside of its mechanical meaning is inappropriate is that “terms and nomenclature used to describe exercise should abide by the *Système International d’Unités* (SI)”.¹ However, the idea that terms and nomenclature used to describe exercise should abide by the SI is incorrect. Operationalising so that the SI can be used is not an obligation and is, in fact, not a requirement of science. To exemplify this, psychological and perceptual variables are commonly measured without the use of the SI, with such measures being common across a range of scientific

areas including psychology, psychophysics, sociology, and exercise and sports science. Accordingly, to consider such measures as ‘unscientific’ because they do not abide by the SI, would be to consider the above-mentioned fields unscientific, which is non-sensical.

In the presented statement by Staunton and colleagues,¹ the pairing of SI units with the verb “describe” highlights a common misunderstanding worth expanding: the conflation of how a construct can be *measured* with how it should be *described*. Indeed, it is the constitutive definition and not the operational definition that can describe a construct. This misconception provides a useful platform for the clarification of some important concepts regarding scientific constructs: the difference between the label, descriptive definition and operational definition of a construct.

3. Understanding constructs: labelling, conceptualisation and operationalisation: from constructs to variables

According to the American Physiological Association dictionary a construct is “a complex idea or concept formed from a synthesis of simpler ideas” (<https://dictionary.apa.org/>). A construct is a mental construction that is differentiated with concepts primarily by the level of abstraction, which is lower for concepts, or in terms of generalisability.^{3,4} Stenner et al⁵ defined a construct as “the means by which science orders observations”. An essential component of the scientific process is to determine a clear description of a construct and how it can be measured.^{6,7} Accordingly, to be considered scientific (i.e., to be investigated using the scientific method), a construct would need to possess at least three characteristics: a construct label, a constitutive definition and operational definitions (Fig. 1).

1. The construct label

Any mental construction of a phenomenon should be identifiable and communicable. For this reason, we must assign a name to this construct: a label. *Training load* is the label assigned to a construct that has been created by observing a phenomenon (training). Although labelling is the process of assigning a descriptive word or phrase to a construct, labels are usually not self-explanatory and can be ambiguous. Accordingly, it is essential to clarify what we mean when we use a label by providing a constitutive definition.

2. Constitutive definition

A constitutive (also called theoretical, conceptual, or descriptive) definition is a description that uses other concepts and gives meaning to the construct or concept under investigation.^{6–9} The conceptualisation of a construct is needed to explain to others the construct identified

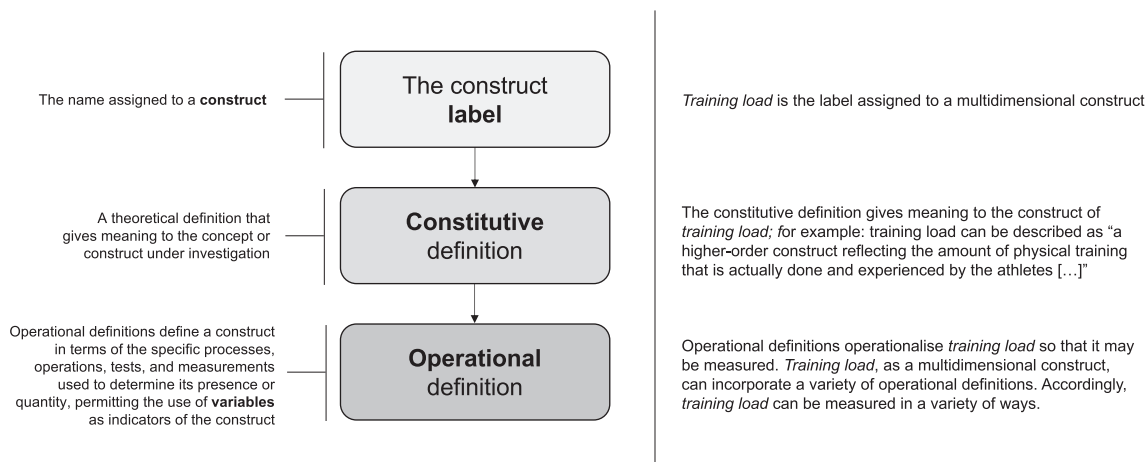


Fig. 1. From construct to variables: process.

by the label. Conceptualisation refers to the mental process by which constructs, and their constituent components, are defined in precise terms. Conceptualisation is an inductive process based on background knowledge. Definitions should be clear and unambiguous, meaning they should allow individuals to identify the main concepts, which are the ones that will be operationalised.

Descriptions should reflect current understanding

While we agree that some definitions are unclear and may be misinterpreted, and indeed, a definition of a higher-order construct should be precise, such a definition cannot be too specific because it should allow for the inclusion of subdimensions. It is therefore important not to conflate the necessity of generic definitions with vagueness. Finding a balance is often challenging. A constitutive definition should reflect *current* conceptions of the construct and related concepts. Importantly, conceptual engineering involves not only the development of new concepts but also the *revision* and *adaptation* of old concepts to a given task and context (concept re-engineering).^{10–12} A model example of concept re-engineering can be found within the work of Knuttgen and Kraemer,¹³ which was referred to but misrepresented within the article by Staunton and colleagues.¹ Unlike Staunton and colleagues,¹ who are calling for the discarding of various terms such as *load* and *training load*, Knuttgen and Kraemer¹³ did not call for the discarding of terms. The goal of Knuttgen and Kraemer¹³ was *not* to “remind[ed] sport and exercise scientists that an isometric muscle ‘contraction’ is not possible”,¹ and that therefore this term needs to be abandoned or replaced. Rather, Knuttgen and Kraemer¹³ accepted that physiologists had used the term *contraction* outside of its dictionary meaning, acknowledging that “the term contraction would appear to be firmly entrenched by history and tradition as referring to activated muscle.” Whilst they made it clear that adopting the dictionary definition of *contraction* (i.e., shortening or shortening and thickening) would create a contradiction when combined with the terms concentric, eccentric, and isometric, their ‘solution to the dilemma’ was exemplary. They proposed to “define muscle contraction as, ‘the active state of muscle tissue; the attempt of muscle to shorten’”, and that “no directionality is to be inferred”.¹³ Accordingly, Knuttgen and Kraemer¹³ did not suggest to abandon the term contraction, rather, they opted to retain and redefine the term contraction, to better fit the context within which it was commonly used.

3. Operational definition

Operational definitions are fundamental to the scientific process. They outline a construct in terms of the specific processes, tests, and measurements used to determine its presence or quantity, permitting the use of measurements as indicators of the construct. Accordingly, by operationalising the concepts used to describe the construct, we create variables reflecting the construct. This notion is well-established in science. In a seminal paper published almost 90 years ago entitled “the operational basis of psychology”, S. Stevens explained: “We must first define an operation; and, if we are to be consistent, we must define it operationally. An operation is a performance which we execute in order to make known a concept.”¹⁴ The American Psychological Association (<https://dictionary.apa.org/>) defines an operational definition as “a description of something in terms of the operations (procedures, actions, or processes) by which it could be observed and measured”.

In other words, once a label has been assigned to a construct, and once a constitutive or descriptive definition is provided, operational definitions are needed to obtain indicators that reflect the construct. It is at this point that the SI is relevant. Accordingly, stating that “terms and nomenclature used to *describe* exercise should abide by the SI” is conceptually flawed. Terms to ‘describe exercise’ should only be conceptualised and defined. Afterward, the concepts used in the description can be operationalised to obtain measures reflecting the concepts, eventually using the SI (if appropriate and necessary). Even within a physics context, as stated by Slisko and Dykstra,¹⁵ “[in physics]

we intend to define a physical quantity by its operational definition, which leads to a specific dimension (and unit) and not vice versa”.

4. Multidimensional constructs and the *training load* construct

An important aspect when conceptualising constructs is understanding whether they are unidimensional or multidimensional. Unidimensional constructs are expected to have a single underlying dimension. They can be measured using a single measure or test, while multidimensional constructs consist of interrelated attributes or dimensions within a multidimensional domain.⁴ *Training load* is a multidimensional construct, acting as a higher-order construct whereby the subdimensions of *external training load* and *internal training load* are different forms manifested by the construct. Considering the underlying dimensions of *training load* (which are also multidimensional constructs with their own description), a single operational definition (i.e., a single measure) cannot encapsulate all possible dimensions of the constructs of *training load*, *external training load*, or *internal training load*. Indeed, it is the constitutive definition and not the operational definition that can describe a multidimensional construct, and that can (conceptually) capture all of the dimensions of the construct. According to their multidimensional nature, various operational definitions may exist, with these definitions permitting a variety of measures reflecting the constructs.

Staunton and colleagues¹ stated that they “are not aware of any other field of science where different constructs are combined into a meta-construct, like ‘training load’.” However, similar (and much more complex) multi-level structures and higher-order constructs are common, especially in other disciplines such as psychology, clinimetrics, and social sciences. For example, the construct of *wellbeing* can include various dimensions such as standard of living, social relationships, and health. Social relations can have two sub-dimensions such as community and personal, and health has other sub-dimensions such as physical and psychological health, and so on.¹⁶ Even in sport and exercise science examples of other multidimensional constructs exist, such as “musculoskeletal fitness”, which includes the sub-dimensions of muscle strength, muscle endurance and muscle power.¹⁷ Similarly to *training load*, each of these sub-dimensions can also have other lower-level dimensions, and all can be operationalised in a variety of ways, resulting in a number of different variables.

As explained in a seminal article by Cronbach & Meehl in 1955,¹⁸ the multi-level structures of multidimensional constructs help to define a nomological network. The classification and differentiations proposed for physical training load are not just part of a taxonomic or purely semantic exercise, but are components of a conceptual framework that can be used to design validation studies, and that supports the research process.¹⁹ The validity of the structure of the multidimensional construct of *training load*, along with its sub-dimensions, to date, can only be inferred by studies demonstrating that measures of internal and external training load are related. For example, the strength of the associations reported in the literature, and the possibility to dissociate internal and external load, support that the sub-dimensions are indeed interrelated but different constructs, coherent with the proposed hierarchy and structure (Fig. 2).²⁰

5. Conclusion

In sport and exercise science, as in other sciences, clarity and accuracy of the relevant specialised language are crucial for disseminating knowledge, reproducibility of results of experiments, and facilitation of meaningful discourses between researchers. Staunton and colleagues¹ have properly identified an unjustified proliferation of the use of various terms and definitions in sport and exercise science. However, we propose some of the main arguments underpinning their framework are conceptually flawed. The implicit allegation that the exercise and sport science community use unscientific terms is unjustified and not supported by

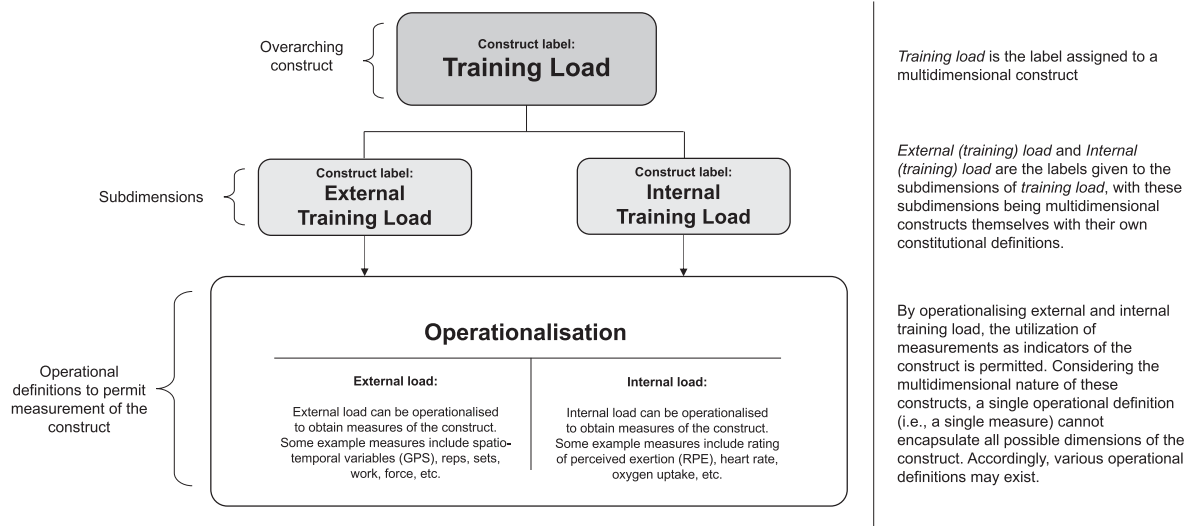


Fig. 2. Multidimensional structure of the training load construct.

an appropriate theoretical explanation. Furthermore, the solutions proposed by Staunton and colleagues¹ are also questionable and contradictory (see details in the preprint: <https://sportrxiv.org/index.php/server/preprint/view/16>). Some fundamental aspects pertaining to construct labelling, conceptualisation, and operationalisation have been missed, while operationalising constructs so that measures of the construct use the SI is not an obligation of science.

What's next?

Awareness of the fundamental issues presented in this article may aid in reducing currently overlapping and competing constructs and definitions, and consequently, lessen confusion and misunderstanding. Researchers and practitioners can continue to use the terms *load* and *training load* as they do not and cannot breach any scientific principles. Ultimately, it is up to the sport and exercise science community to provide definitions of its shared constructs.

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References

1. Staunton CA, Abt G, Weaving D et al. Misuse of the term 'load' in sport and exercise science. *J Sci Med Sport* 2021;S1440–2440(21):00212–00217.
2. MacDonald DA, Friedman HL, Brewczynski J et al. Spirituality as a scientific construct: testing its universality across cultures and languages. *PLoS One* 2015;10(3):e0117701.
3. Markus KA. Constructs, concepts and the worlds of possibility: connecting the measurement, manipulation, and meaning of variables. *Meas Interdiscip Res Perspect* 2008;6(1–2):54–77.
4. Bhattacharjee. *Social Science Research: Principles, Methods and Practices*, 2012.
5. Stenner AJ, Smith M, Burdick DS. Toward a theory of construct definition. *J Educ Meas* 1983;20(4):305–316.
6. Reznick JS. Defining constructs and variables. In: Reznick JS, ed. *The Developmental Scientist's Companion: Improving Research Methodology and Achieving Professional Success*, Cambridge, Cambridge University Press, 2017.
7. Hoy WK. Concepts, variables, and research problems. *Quantitative Research in Education: A Primer*, Thousand Oaks, California, SAGE Publications, Inc., 2010.
8. Middendorp CP. On the conceptualization of theoretical constructs. *Qual Quant* 1991;25(3):235–252.
9. Willer D, Webster M. Theoretical concepts and observables. *Am Sociol Rev* 1970;35(4):748–757.
10. Vessonen E. Conceptual engineering and operationalism in psychology. *Synthese* 2021.
11. Chalmers DJ. What is conceptual engineering and what should it be? *Inquiry* 2020;1–18.
12. Cappelen H. Fixing language. An essay on conceptual engineering. *Z Philos Forsch* 2019;73(1):169–173.
13. Knuttgen HG, Kraemer WJ. Terminology and measurement in exercise performance. *J Strength Cond Res* 1987;1(1):1–10.
14. Stevens SS. The operational basis of psychology. *Am J Psychol* 1935;47(2):323–330.
15. Sljako J, Dykstra DI. The role of scientific terminology in research and teaching: is something important missing? *J Res Sci Teach* 1997;34:655–660.
16. Olsen JA, Misajon R. A conceptual map of health-related quality of life dimensions: key lessons for a new instrument. *Qual Life Res* 2020;29(3):733–743.
17. Lang JJ, Smith JJ, Tomkinson GR. Global surveillance of cardiorespiratory and musculoskeletal fitness. In: Brusseau TA, Fairclough SJ, Lubans DR, eds. *The Routledge Handbook of Youth Physical Activity*, Abingdon, Routledge, 2020.
18. Cronbach LJ, Meehl PE. Construct validity in psychological tests. *Psychol Bull* 1955;52(4):281–302.
19. Jeffries AC, Marcora SM, Coutts AJ et al. Development of a revised conceptual framework of physical training for use in research and practice. *Sports Med* 2021.
20. McLaren SJ, Macpherson TW, Coutts AJ et al. The relationships between internal and external measures of training load and intensity in team sports: a meta-analysis. *Sports Med* 2018;48(3):641–658.